



Report 6 - Map

October 3, 2026

0.1 Introduction

3 little kernels to implement.

0.2 grayscale image binarization

```
@cuda.jit
def binarization(src, dst, threshold):
    # where are we in the input?
    x = cuda.threadIdx.x + cuda.blockIdx.x * cuda.blockDim.x
    y = cuda.threadIdx.y + cuda.blockIdx.y * cuda.blockDim.y

    if y < src.shape[0] and x < src.shape[1]:
        src[y, x] = [255 if src[y, x] > threshold else 0]
```

0.3 brightness control

This one takes an additional argument and will brighten or darken the image based on its value.

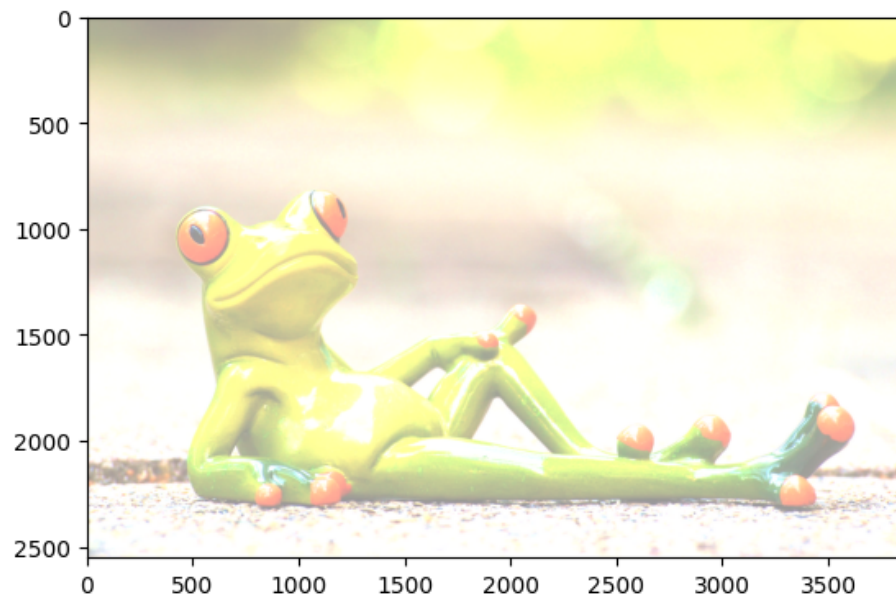
```
@cuda.jit
def brightness_control(src, dst, value):
    # where are we in the input?
    x = cuda.threadIdx.x + cuda.blockIdx.x * cuda.blockDim.x
    y = cuda.threadIdx.y + cuda.blockIdx.y * cuda.blockDim.y
    if y < src.shape[0] and x < src.shape[1]:
        for i in range(3):
            dst[y, x, i] = min(255, (max(0, src[y, x, i] + value)))
```

Example: frog with +150 brightness

0.4 blending two images

This one take a coefficient as an argument and will blend the 2 images given. The coefficient acts as the weight for the first image.

```
@cuda.jit
def blend_images(src1, src2, q, dst):
    # where are we in the input?
    x = cuda.threadIdx.x + cuda.blockIdx.x * cuda.blockDim.x
    y = cuda.threadIdx.y + cuda.blockIdx.y * cuda.blockDim.y
    if y < src1.shape[0] and x < src1.shape[1]:
        for i in range(3):
            dst[y][x][i] = q * src1[y][x][i] + (1 - q) * src2[y][x][i]
```



Example: the frog has a 0.6 coefficient, meaning the blobfish has 0.4

